



COMPUTER NETWORKS

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Transport Layer

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Transport Layer - Roadmap

3.1 Transport-layer Services

3.2 Multiplexing and Demultiplexing

3.3 Connectionless Transport: UDP

Transport Layer - Roadmap

3.1 Transport-layer Services

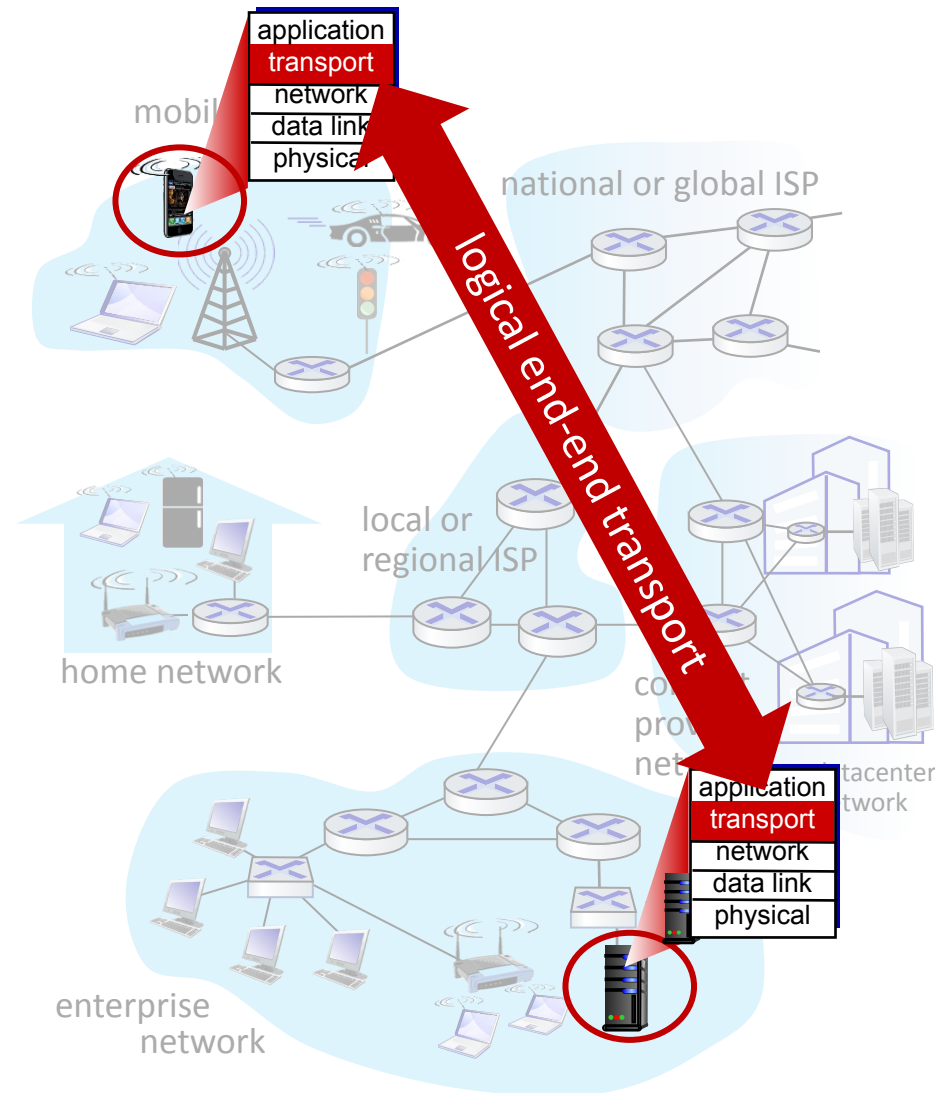
3.2 Multiplexing and Demultiplexing

3.3 Connectionless Transport: UDP

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Transport Services and Protocols

- provide *logical communication* between application processes running on different hosts
- transport protocols actions in end systems:
 - sender: breaks application messages into *segments*, passes to network layer
 - receiver: reassembles segments into messages, passes to application layer
- two transport protocols available to Internet applications
 - TCP, UDP





household analogy:

*12 kids in Ann's house
sending letters to 12 kids
in Bill's house:*

- **hosts** = houses
- **processes** = kids
- **app messages** = letters in envelopes

- **network layer:** logical communication between *hosts*
- **transport layer:** logical communication between *processes*
 - relies on, enhances, network layer services

household analogy:

*12 kids in Ann's house
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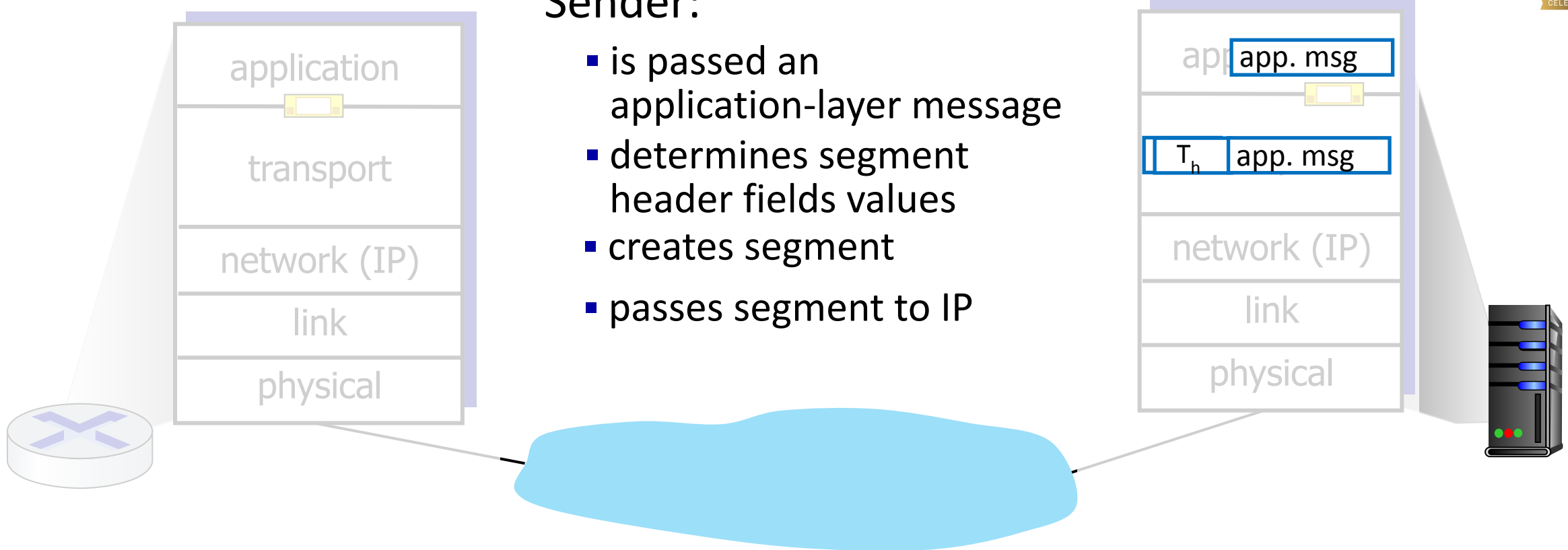
- **hosts** = houses
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Transport Layer Actions

Sender:

- is passed an application-layer message
- determines segment header fields values
- creates segment
- passes segment to IP

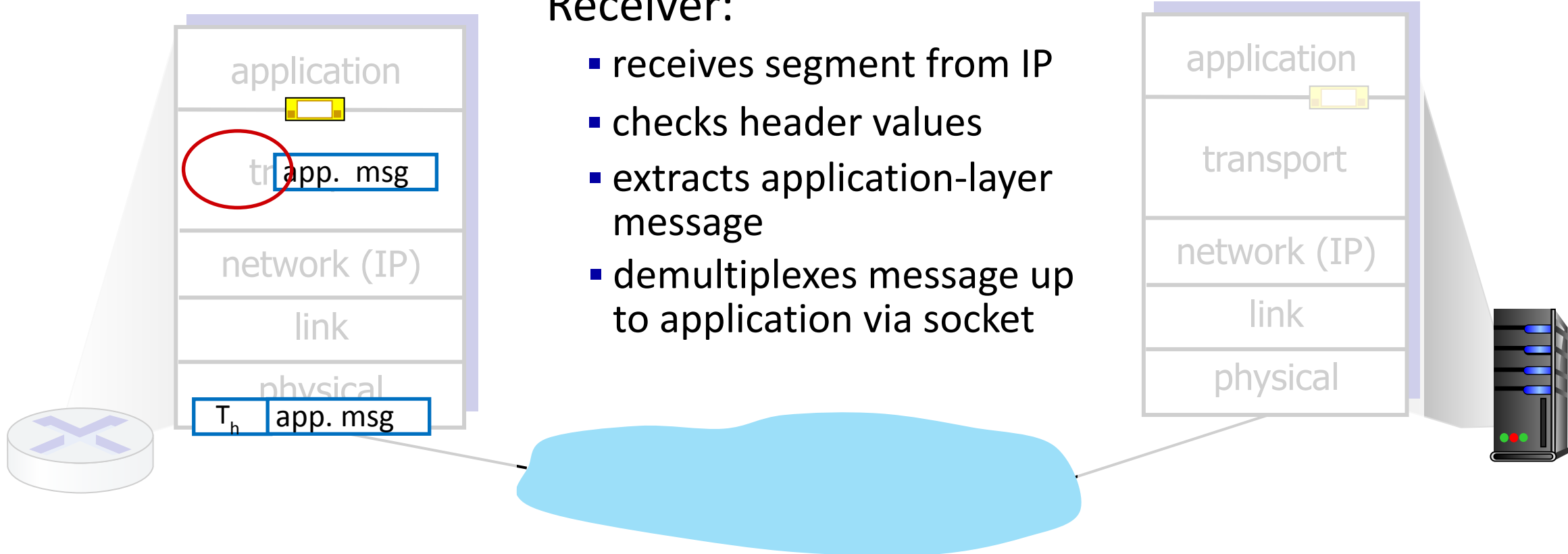


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Transport Layer Actions

Receiver:

- receives segment from IP
- checks header values
- extracts application-layer message
- demultiplexes message up to application via socket

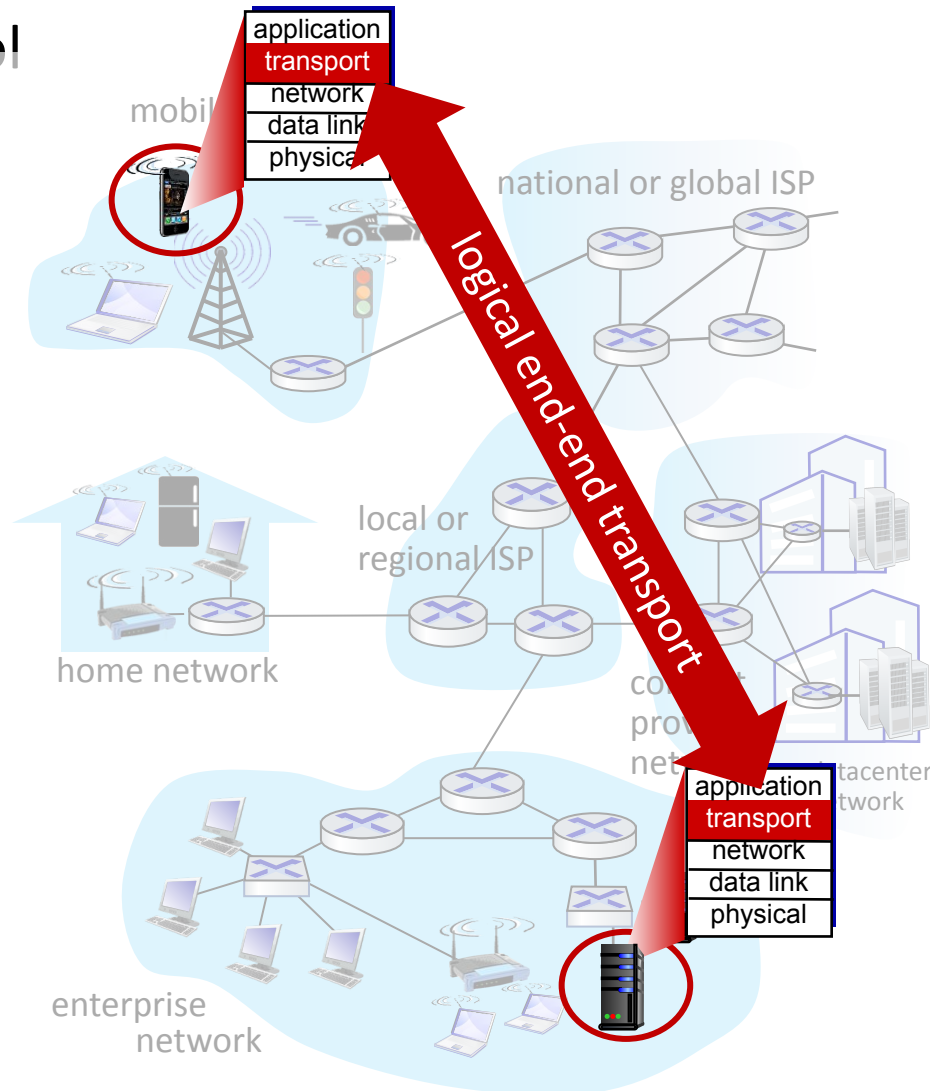


- **TCP:** Transmission Control Protocol

- reliable, connection oriented
- in-order delivery
- congestion control
- flow control
- connection setup

- **UDP:** User Datagram Protocol

- unreliable, connectionless
- unordered delivery
- no-frills extension of “best-effort” IP
- services not available:
 - delay guarantees
 - bandwidth guarantees



- Transport Layer – Explained – <https://youtu.be/FxFJ1XlWtdI>
- Transport Layer Services – IIT Kharagpur – <https://youtu.be/8-3CSAksYU>
- Transport Layer – Process to Process Delivery – <https://youtu.be/9e4vTcaEYCg>



Thank You
For Your Attention



THANK YOU

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